

GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY

NEW DELHI

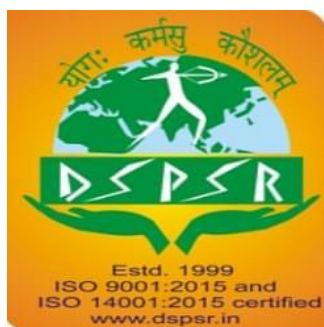


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INFORMATION SYSTEM MANAGEMENT

Submitted in partial fulfilment of the requirement for the award of the degree of

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Submitted to:

MRS. PRINCY AGGARWAL

(Assistant Professor)

Submitted by:

Sunny pawar

BBA 5B

04221701722

CERTIFICATE

This is to certify that this ISM Lab Practical File is an academic work done by Ms. Sunny pawar submitted in the partial fulfilment of the requirements for the award of degree of bachelors of business administration at Delhi School of Professional Studies and Research, New Delhi under my guidance and direction.

Ms. Sunny pawar has given an undertaking that the information presented in the project has not been submitted earlier.

MS. PRINCY AGGARWAL
(ASSISSTANT PROFESSOR)

DECLARATION

I take this opportunity to express my profound gratitude and deep regards to my guide Ms. Princy Agarwal for this exemplary guidance, monitoring and constant encouragement throughout the course of the project. The blessing, help and guidance given by her time to time shall carry me a long way in the journey of life on which I am about to embark.

Last but not the least, my sincere thanks to my parents and friends for their wholehearted support and encouragement.

I also hereby declare that the ISM Lab Practical File under the guidance of Ms. Princy Agarwal is my original work and it has not been submitted earlier in any other university or institution.

Sunny pawar
(BBA 5th Semester)
Enrollment No.- 00421701722

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QUESTION 1: INTRODUCTION TO SQL. EXPLAIN WHAT ARE DIFFERENT

TYPES OF DATA TYPES IN SQL.

SQL (Structured Query Language) is a domain-specific language used in programming and designed for managing data held in a relational database management system (RDBMS), or for stream processing in a relational data stream management system (RDSMS). It is particularly useful in handling structured data, i.e. data incorporating relations among entities and variables.

SQL (Structured Query Language) is a database sub-language for querying and modifying relational databases. It was developed by IBM Research in the mid 70's and standardized by ANSI (American National Standards Institutes) in 1986.

The Structured Query Language (SQL) is a language of databases. All modern relational databases, including Access, File-Maker pro, Microsoft SQL Server and Oracle use SQL as their basic building block. All of the graphical user interfaces that provide data entry and manipulation functionality are nothing more than SQL translators. They take the actions you perform graphically and convert them to SQL commands understood by the database. Using SQL, you can insert records, update records, and delete records. You can also create new database objects such as databases and tables. And you can drop (delete) them. SQL offers two main advantages over older read-write APIs such as ISAM or VSAM. Firstly, it introduced the concept of accessing many records with one single command. Secondly, it eliminates the need to specify how to reach a record, e.g. with or without an index. Originally based upon relational algebra and tuple relational calculus, SQL consists of many types of statements, which may be informally classed as sub languages, commonly: a data query language (DQL), a data definition language (DDL), a data control language (DCL), and a data manipulation language (DML). The scope of SQL includes data query, data manipulation (insert, update and delete), data definition (schema creation and modification), and data

access control. Although SQL is essentially a declarative language (4GL), it includes also procedural elements.

SQL was one of the first commercial languages to utilize Edgar Codd's relational model. The model was described in his influential 1970 paper, "A Relational Model of Data for Large Shared Data Banks". Despite not entirely adhering to the relational model as described by Codd, it became the most widely used database language.

HISTORY

SQL was initially developed at IBM by Donald D. Chamberlin and Raymond F. Boyce after learning about the relational model from Ted Codd in the early 1970s. This version, initially called SEQUEL (Structured English Query Language), was designed to manipulate and retrieve data stored in IBM's original quasi-relational database management system, System R, which a group at IBM San Jose Research Laboratory had developed during the 1970s.

Chamberlin and Boyce's first attempt of a relational database language was Square, but it was difficult to use due to subscript notation. After moving to the San Jose Research Laboratory in 1973, they began work on SEQUEL. The acronym SEQUEL was later changed to SQL because "SEQUEL" was a trademark of the UK-based Hawker Siddeley Dynamics Engineering Limited company. After testing SQL at customer test sites to determine the usefulness and practicality of the system, IBM began developing commercial products based on their System R prototype including System/38, SQL/DS, and DB2, which were commercially available in 1979, 1981, and 1983, respectively.

In the late 1970s, Relational Software, Inc. (now Oracle Corporation) saw the potential of the concepts described by Codd, Chamberlin, and Boyce, and developed their own SQL-based RDMS with aspirations of selling it to the U.S.

Navy, Central Intelligence Agency, and other U.S. government agencies. In June 1979, Relational Software, Inc. introduced the first

commercially available implementation of SQL, Oracle V2 (Version2) for VAX computers.

DESIGN

SQL deviates in several ways from its theoretical foundation, the relational model and its tuple calculus. In that model, a table is a set of tuples, while in SQL, tables and query results are lists of rows: the same row may occur multiple times, and the order of rows can be employed in queries (e.g. in the LIMIT clause). Critics argue that SQL should be replaced with a language that returns strictly to the original foundation: for example, see The Third Manifesto. However, no known proof exists that such uniqueness cannot be added to SQL itself, or at least a variation of SQL. In other words, it's quite possible that SQL can be "fixed" or at least improved in this regard such that the industry may not have to switch to a completely different query language to obtain uniqueness. Debate on this remains open.

WHAT CAN SQL DO?

SQL can execute queries against a database. SQL can retrieve data from a database. SQL can insert records in a database. SQL can update records in a database. SQL can delete records from a database.

SQL can create new databases.

SQL can create new tables in a database.

SQL can create stored procedures in a database. SQL can create views in a database.

SQL can set permissions on tables, procedures, and views.

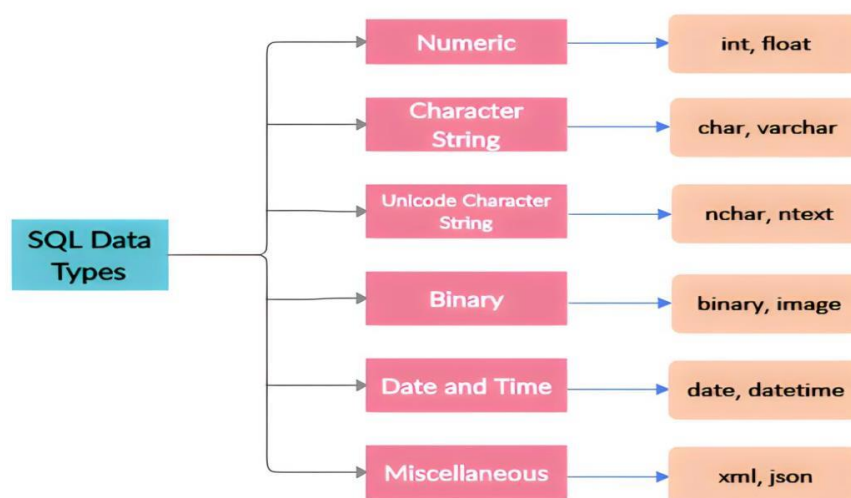
DIFFERENT DATA TYPES IN SQL

SQL Data Types define the type of value that can be stored in a table column. For example, if we want a column to store only integer values, then we can define its data type as int.

SQL DATA TYPES

SQL data types can be broadly divided into following categories.

- 1.Numeric data types such as int, tinyint, bigint, float, real etc.
- 2.Date and Time data types such as Date, Time, Datetime etc.
- 3.Character and String data types such as char, varchar, text etc.
- 4.Unicode character string data types, for example nchar, nvarchar, ntext etc.
- 5.Binary data types such as binary, varbinary etc.
- 6.Miscellaneous data types – clob, blob, xml, cursor, table etc



SQL DATA TYPES IMPORTANT POINTS

- Not all data types are supported by every relational database vendor. For example, Oracle database doesn't support DATETIME and MySQL doesn't support CLOB data type. So, while designing database schema and writing sql queries, make sure to check if the data types are supported or not.
- Data types listed here doesn't include all the data types, these are the most popularly used data types. Some relational database vendors have their own data types that might be not listed here. For example, Microsoft SQL Server has data types but since it's not supported by other popular database vendors, it's not listed here.

- Every relational database vendor has its own maximum size limit for different data types, you don't need to remember the limit. Idea is to have the knowledge of what data type to be used in a specific scenario.

THE FOLLOWING TABLE LISTS THE GENERAL DATA TYPES IN SQL: -

CHARACTER(n)- Character string. Fixed-length n

VARCHAR(n) OR CHARACTER VARYING(n)- Character string. Variable length. Maximum length n

BINARY(n)- Binary string. Fixed-length n

BOOLEAN- Stores TRUE or FALSE values

VARBINARY(n) OR BINARY VARYING(n)- Binary string. Variable length. Maximum length n

INTEGER(p)-Integer numerical (no decimal). Precision

SMALLINT- Integer numerical (no decimal). Precision

BIGINT- Integer numerical (no decimal). Precision

The diagram illustrates a table structure. At the top, the word "Columns" is written in green. Three downward-pointing arrows indicate the columns: "EMPLOYEE_ID", "EMPLOYEE_NAME", and "COUNTRY". On the left, the word "Rows" is written in green. Three rightward-pointing arrows indicate the rows. The table contains three rows of data.

EMPLOYEE_ID	EMPLOYEE_NAME	COUNTRY
123	Manu Manjunatha	India
124	Advith	India
125	Likitha	India

QUESTION 2: EXPLAIN SQL DDL COMMANDS WITH SYNTAX AND EXAMPLES.

The Data Definition Language (DDL) is used to create and destroy databases and databases objects. These SQL

statements define the structure of a database, including rows, columns, tables, indexes, and databases specifics such as file locations.

- 1. CREATE** – To create objects in the database.
- 2. ALTER** – Alters the structure of the database.
- 3. DROP** – Delete objects from the databases.
- 4. TRUNCATE**–is used to remove all records from a table, including all spaces allocated for the records are removed.
- 5. COMMENT** –is used to add comments to the data dictionary.
- 6. RENAME** –is used to rename an object existing in the database.

THE CREATE TABLE COMMAND: -

The create table command defines each column of the table uniquely. Each column has minimum of three attributes.

1. Name
2. Data type
3. Size (column width).

Each table column definition is a single clause in the create table syntax. Each table column definition is separated from the other by a comma. Finally, the SQL statement is terminated with a semicolon. The Structure of Create Table Command

TABLE NAME IS STUDENT

Column name	Data type	Size
Reg_no	varchar2	10
Name	Char	30
DOB	Date	
Address	varchar2	50

EXAMPLE: - CREATE TABLE Student (Reg_no varchar2(10), Name char (30), DOB date, Address varchar2(50));

The DROP COMMAND: -

SYNTAX: DROP TABLE <table_name>

EXAMPLE: DROP TABLE Student; It will destroy the table and all data which will be recorded in it.

THE TRUNCATE COMMAND: -

SYNTAX: TRUNCATE TABLE <Table_name>

EXAMPLE: TRUNCATE TABLE Student;

The RENAME COMMAND: -

SYNTAX: RENAME <OldTableName> TO <NewTableName>

EXAMPLE: RENAME <Student> TO <Stu>

The old name table was **Student** now new name is the **Stu**. **The ALTER TABLE COMMAND: -**

By The use of ALTER TABLE Command, we can **modify** our exiting table.

1. Adding New Columns

SYNTAX: ALTER TABLE <table_name>

ADD (<NewColumnName> <Data_Type>(<size>),n)

EXAMPLE: ALTER TABLE Student ADD (Age number (2), Marks number (3)); The student table is already existing and then we added

two more columns **Age** and **Marks** respectively, by the use of above command.

2 Dropping a Column from the Table SYNTAX: ALTER TABLE <table_name> DROP COLUMN <column_name>

EXAMPLE: ALTER TABLE Student DROP COLUMN Age This command will drop particular column

3 Modifying Existing Table

SYNTAX: ALTER TABLE <table_name> MODIFY (<column_name> <NewDataType>(<NewSize>))

EXAMPLE:

ALTER TABLE Student MODIFY (Name Varchar2(40));

The Name column already exist in Student table, it was char and size 30, now it is modified by Varchar2 and size 40.

4 Restriction on the ALTER TABLE

Using the ALTER TABLE clause the following tasks cannot be performed. Change the name of the table Change the name of the column Decrease the size of a column if table data exists

QUESTION 3: EXPLAIN SQL DML COMMAND WITH SYNTAX &

EXAMPLE. The SQL commands that deals with the manipulation of data present in database belong to DML or Data Manipulation Language and this includes most of the SQL statements.

EXAMPLE OF DML:

SELECT – is used to retrieve data from a database.

INSERT – is used to insert data into a table.

UPDATE – is used to update existing data within a table.

DELETE – is used to delete records from a database table.

THE INSERT COMMAND: -SQL INSERT statement allows to insert single or multiple records into the table.

SYNTAX: INSERT INTO table

(column-1, column-2, column-n) (value-1, value-2, ... value-n);

THE UPDATE COMMAND: -

UPDATE QUERY is used to update existing records in the table

SYNTAX: UPDATE table SET column1=value1, column2=value2...
WHERE condition;

EXAMPLE: The following example update the Mike's salary to 35000 in the Employee table. UPDATE Employee SET Salary= 35000
WHERE

Name = 'Mike';

THE DELETE COMMAND:

DELETE QUERY is used to delete selected rows, or all rows from the table.

SYNTAX: DELETE from table WHERE condition;
EXAMPLE: The following example delete the record where EID equals to 10 from the Employee table, DELETE from Employee WHERE EID=10;

QUESTION4: CREATE A TABLE NAMED CUSTOMER (C_ID, C_NAME, ADDRESS, CITY, PINCODE, COUNTRY). INSERT AT LEAST 10 VALUES, DISPLAY THE TABLE.

- a. Create database students;
 - b. Use students;
 - c. Create table customer (customer_id int, customer_name varchar (20), address varchar (50), city varchar (20), pin code int, country varchar (20));
1. Insert Into customer value(01, 'Mayank walia', '9/11 Uttam nagar', 'New delhi', 110045, 'India');
 2. Insert Into customer value(02, 'Manitosh pradhan', 'b/135 paschim vihar', 'New delhi', 110041, 'India');
 3. Insert Into customer value(03, 'Naman jain', '420 nala supara', 'New delhi', 110001, 'India');
 4. Insert Into customer value(04, 'Shyam bansal', '786 anand vihar', 'New delhi', 110040, 'India');
 5. Insert Into customer value(05, 'Salmon bhai', '92/11 rithala village', 'New delhi', 110068, 'India');
 6. Insert Into customer value(06, 'Vansh jain', '420/2 rohini', 'New delhi', 110085, 'India');
 7. Insert Into customer value(07, 'Rohan massey', '420/4 rohini sec-3', 'New delhi', 110065, 'India');
 8. Insert Into customer value(08, 'Shoeb ansari', '420/9 karol bagh', 'New delhi', 110093, 'India');
 9. Insert Into customer value(09, 'Appu dass', '99A Cr park', 'New delhi', 110099, 'India');
 10. Insert into customer value(10, 'Ansh malik', '9/2/11 kachra vihar', 'New delhi', 110020, 'India');
 11. Select * from customer;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 40
Server version: 5.1.44-community MySQL Community Server (GPL)

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE STUDENTS;
Query OK, 1 row affected (0.00 sec)

mysql> USE STUDENTS;
Database changed
mysql> CREATE TABLE CUSTOMER (CUSTOMER_ID INT, CUSTOMER_NAME VARCHAR(20), ADDRESS VARCHAR(50), CITY VARCHAR(20), PINCODE INT, COUNTRY VARCHAR(20));
Query OK, 0 rows affected (0.17 sec)

mysql> INSERT INTO CUSTOMER VALUE(01,'MAYANK WALIA','9/91 UTTAM NAGAR','NEW DELHI',110045,'INDIA');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO CUSTOMER VALUE(02,'MANITOSH PRADHAN','B/135 PASCHIM VIHAR','NEW DELHI',110041,'INDIA');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO CUSTOMER VALUE(03,'NAMAN JAIN','420 NALA SUPARA','NEW DELHI',110001,'INDIA');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO CUSTOMER VALUE(04,'SHYAM BANSAL','786 ANAND VIHAR','NEW DELHI',110040,'INDIA');
Query OK, 1 row affected (0.05 sec)

mysql> INSERT INTO CUSTOMER VALUE(05,'SALMON BHAI','92/11 RITHALA VILLAGE','NEW DELHI',110068,'INDIA');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO CUSTOMER VALUE(06,'VANSH JAIN','420/2 ROHINI','NEW DELHI',110085,'INDIA');
Query OK, 1 row affected (0.12 sec)

mysql> INSERT INTO CUSTOMER VALUE(07,'ROHAN MASSEY','420/4 ROHINI SEC-3','NEW DELHI',110065,'INDIA');
Query OK, 1 row affected (0.37 sec)

mysql> INSERT INTO CUSTOMER VALUE(08,'SHOEB ANSARI','420/9 KAROL BAGH','NEW DELHI',110093,'INDIA');
Query OK, 1 row affected (0.36 sec)

mysql> INSERT INTO CUSTOMER VALUE(09,'APPU DASS','99A CR PARK','NEW DELHI',110099,'INDIA');
Query OK, 1 row affected (0.38 sec)

mysql> INSERT INTO CUSTOMER VALUE(10,'ANSH MALIK','9/2/11 KACHRA VIHAR','NEW DELHI',110420,'INDIA');
Query OK, 1 row affected (0.10 sec)
```

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT * FROM CUSTOMER;
+-----+-----+-----+-----+-----+-----+
| CUSTOMER_ID | CUSTOMER_NAME | ADDRESS | CITY | PINCODE | COUNTRY |
+-----+-----+-----+-----+-----+-----+
| 1 | MAYANK WALIA | 9/91 UTTAM NAGAR | NEW DELHI | 110045 | INDIA |
| 2 | MANITOSH PRADHAN | B/135 PASCHIM VIHAR | NEW DELHI | 110041 | INDIA |
| 3 | NAMAN JAIN | 420 NALA SUPARA | NEW DELHI | 110001 | INDIA |
| 4 | SHYAM BANSAL | 786 ANAND VIHAR | NEW DELHI | 110040 | INDIA |
| 5 | SALMON BHAI | 92/11 RITHALA VILLAGE | NEW DELHI | 110068 | INDIA |
| 6 | VANSH JAIN | 420/2 ROHINI | NEW DELHI | 110085 | INDIA |
| 7 | ROHAN MASSEY | 420/4 ROHINI SEC-3 | NEW DELHI | 110065 | INDIA |
| 8 | SHOEB ANSARI | 420/9 KAROL BAGH | NEW DELHI | 110093 | INDIA |
| 9 | APPU DASS | 99A CR PARK | NEW DELHI | 110099 | INDIA |
| 10 | ANSH MALIK | 9/2/11 KACHRA VIHAR | NEW DELHI | 110420 | INDIA |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.36 sec)

mysql>
```

QUESTION 5: CREATE TABLE NAMED EMPLOYEE CONTANING

COLOUM (emp_id, emp_name, emp_desig, DOB, emp_sal, emp_dept). Insert atleast 10 values. Constraints applied on Employee table:

- 1) Emp_id should be primary key
 - 2) Emp_sal should not contain any Null Value
 - 3) Emp_desig should be Unique
- a. Create database employees;
 - b. Use employees;
 - c. Create table employee (emp_id int primary key, emp_name varchar (20), emp_desig varchar (30) unique, DOB date, emp_sal int not null, emp_dept varchar(50));
 - a. Insert Into employee values(01,'AM', 'sales manager', '1980/03/04', 10000, 'sales');
 - b. Insert Into employee values(02, 'RM', 'marketing manager','1990/09/03', 20000, 'marketing');
 - c. Insert Into employee values(03, 'SA', 'research manager', '2000/03/04', 30000, 'research');
 - d. Insert Into employee values(04,'SS','production manager', '2001/04/05', 40000, 'production');
 - e. Insert Into employee values(05,'VJ','purchase manager', '1980/02/07', 50000, 'purchase');
 - f. Insert Into employee values(06, 'HS', 'promotion manager', '2002/05/7', 60000, 'promotion');
 - g. Insert Into employee values(07,'NJ','investment manager', '2003/05/09', 70000, 'investment');

h. Insert Into employee

values(08,'MW','operationsmanager','2004/01/04',80000,
'operational');

i. Insert Into employee values(09,'SB','finance manager','2001/02/02',
90000, 'finance');

j. Insert Into employee values(10,'MP','HR manager','2001/09/09',
100000, 'HR');

Select * from employee;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 42
Server version: 5.1.44-community MySQL Community Server (GPL)

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE EMPLOYEES;
Query OK, 1 row affected (0.00 sec)

mysql> USE EMPLOYEES;
Database changed
mysql> CREATE TABLE EMPLOYEE(EMP_ID INT PRIMARY KEY, EMP_NAME VARCHAR(20), EMP_DESIGN VARCHAR(30) UNIQUE, DOB DATE, EMP_SAL INT NOT NULL, EMP_DEPT VARCHAR(50));
Query OK, 0 rows affected (0.27 sec)

mysql> INSERT INTO EMPLOYEE VALUES(01,'AM','SALES MANAGER','1980/03/04',10000,'SALES');
Query OK, 1 row affected (0.30 sec)

mysql> INSERT INTO EMPLOYEE VALUES(02,'RM','MARKETING MANAGER','1990/09/03',20000,'MARKETING');
Query OK, 1 row affected (0.05 sec)

mysql> INSERT INTO EMPLOYEE VALUES(03,'SA','RESEARCH MANAGER','2000/03/04',30000,'RESEARCH');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO EMPLOYEE VALUES(04,'SS','PRODUCTION MANAGER','2001/04/05',40000,'PRODUCTION');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO EMPLOYEE VALUES(05,'VJ','PURCHASE MANAGER','1980/02/07',50000,'PURCHASE');
Query OK, 1 row affected (0.36 sec)

mysql> INSERT INTO EMPLOYEE VALUES(06,'HS','PROMOTION MANAGER','2002/05/07',60000,'PROMOTION');
Query OK, 1 row affected (0.36 sec)

mysql> INSERT INTO EMPLOYEE VALUES(07,'NJ','INVESTMENT MANAGER','2003/05/09',70000,'INVESTMENT');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO EMPLOYEE VALUES(08,'MW','OPERATIONS MANAGER','2004/01/04',80000,'OPERATIONAL');
Query OK, 1 row affected (0.37 sec)

mysql> INSERT INTO EMPLOYEE VALUES(09,'SB','FINANCE MANAGER','2001/02/02',90000,'FINANCE');
Query OK, 1 row affected (0.35 sec)

mysql> INSERT INTO EMPLOYEE VALUES(10,'MP','HR MANAGER','2001/09/09',100000,'HR');
```

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT * FROM EMPLOYEE;
+-----+-----+-----+-----+-----+-----+
| EMP_ID | EMP_NAME | EMP_DESIGN | DOB | EMP_SAL | EMP_DEPT |
+-----+-----+-----+-----+-----+-----+
| 1 | AM | SALES MANAGER | 1980-03-04 | 10000 | SALES |
| 2 | RM | MARKETING MANAGER | 1990-09-03 | 20000 | MARKETING |
| 3 | SA | RESEARCH MANAGER | 2000-03-04 | 30000 | RESEARCH |
| 4 | SS | PRODUCTION MANAGER | 2001-04-05 | 40000 | PRODUCTION |
| 5 | VJ | PURCHASE MANAGER | 1980-02-07 | 50000 | PURCHASE |
| 6 | HS | PROMOTION MANAGER | 2002-05-07 | 60000 | PROMOTION |
| 7 | NJ | INVESTMENT MANAGER | 2003-05-09 | 70000 | INVESTMENT |
| 8 | MW | OPERATIONS MANAGER | 2004-01-04 | 80000 | OPERATIONAL |
| 9 | SB | FINANCE MANAGER | 2001-02-02 | 90000 | FINANCE |
| 10 | MP | HR MANAGER | 2001-09-09 | 100000 | HR |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
```

QUESTION 6: CREATE EMPLOYEE TABLE: EID,ENAME, DESG, BRANCH, SALARY, ADDRESS. Insert 10 Records. Salary Column should not contain any Null value ,Salary should be between 5000 to 350000, ENAME should be Unique Perform various SQL command

- 1. Add column in above table date of joining, Experience.**
- 2. Display the Table**
- 3. Complete the Table Definition**
- 4. Delete the Column DESG**
- 5. Find out details of employee whose salary is above 25000.**
- 6. Find out details of employee order by salary.**
- 7. Calculate total no. of records in employee table.**
- 8. List Employees whose salary is between 10000 and 30000.**

1. Create database emp;
2. Use emp;
3. Create table employee(eid int, ename varchar(20) unique, desg varchar(30), branch varchar(50), salary int not null check (salary<350000 and salary>5000), address varchar(50));
 - a) Insert Into employee values(01, 'Vansh jain', 'Factory clerk', 'Rohini west', 10000, '101/1');
 - b) Insert Into employee values(02, 'Sameer salmani', 'Factory labour', 'Rithala village', 20000, '102/2');
 - c) Insert Into employee values(03, 'Rohan massey', 'Factory peon', 'Rohini east', 30000, '103/3');
 - d) Insert Into employee values(04, 'Rahal kumar', 'Factory head', 'Nihal vihar', 40000, '104/4');
 - e) Insert Into employee values(05, 'Om kapoor', 'HR manager', 'Sadar bazaar', 50000, '105/5');

- f) Insert Into employee values(06, 'Salman khan', 'Finance manager', 'Kirti nagar', 60000, '106/6');
- g) Insert Into employee values(07, 'Rajnikant', 'Assistant manager', 'Hauskhas', 70000, '107/7');
- h) Insert Into employee values(08, 'Shahid kapoor', 'Factory labour', 'Madipur', 8000, '108/8');
- i) Insert Into employee values(09, 'Amir khan', 'Factory watchman', 'Karampura', 9000, '109/9');
- j) Insert Into employee values(10, 'Sharuk khan', 'HR manager', 'Paschim vihar', 100000, '110/10');

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 78
Server version: 5.1.44-community MySQL Community Server (GPL)

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE E_MPLYEE;
Query OK, 1 row affected (0.00 sec)

mysql> USE E_MPLYEE;
Database changed
mysql> CREATE TABLE EMPLOYEE(ID INT, NAME VARCHAR(20) UNIQUE, DESG VARCHAR(30), BRANCH VARCHAR(50), SALARY INT NOT NULL
CHECK (SALARY<350000 AND SALARY>5000), ADDRESS VARCHAR(50));
Query OK, 0 rows affected (0.16 sec)

mysql> INSERT INTO EMPLOYEE VALUES(01, 'VANSH JAIN', 'FACTORY CLERK', 'ROHINI WEST', 10000, '101/1');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO EMPLOYEE VALUES(02, 'SAMEER SALMANI', 'FACTORY LABOUR', 'RITHALA', 20000, '102/2');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO EMPLOYEE VALUES(03, 'ROHAN MASSEY', 'FACTORY PEON', 'ROHINI EAST', 30000, '103/3');
Query OK, 1 row affected (0.14 sec)

mysql> INSERT INTO EMPLOYEE VALUES(04, 'RAHUL KUMAR', 'FACTORY HEAD', 'NIHAL VIHAR', 40000, '104/4');
Query OK, 1 row affected (0.13 sec)

mysql> INSERT INTO EMPLOYEE VALUES(05, 'OM KAPOOR', 'HR MANAGER', 'SADAR BAZAAR', 50000, '105/5');
Query OK, 1 row affected (0.13 sec)

mysql> INSERT INTO EMPLOYEE VALUES(06, 'SALMAN KHAN', 'FINANCE MANAGER', 'KIRTI NAGAR', 60000, '106/6');
Query OK, 1 row affected (0.13 sec)

mysql> INSERT INTO EMPLOYEE VALUES(07, 'RAJNIKANT', 'ASSISTANT MANAGER', 'HAUSKHAS', 70000, '107/7');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO EMPLOYEE VALUES(08, 'SAHID KAPOOR', 'FACTORY LABOUR', 'MADIPUR', 8000, '108/8');
Query OK, 1 row affected (0.36 sec)

mysql> INSERT INTO EMPLOYEE VALUES(09, 'AMIR KHAN', 'FACTORY WATCHMAN', 'KARAMPURA', 9000, '109/9');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO EMPLOYEE VALUES(10, 'SHARUK KHAN', 'HR MANAGER', 'PASCHIM VIHAR', 100000, '110/10');
```

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT * FROM EMPLOYEE;
+----+-----+-----+-----+-----+-----+
| ID | NAME      | DESG      | BRANCH    | SALARY | ADDRESS |
+----+-----+-----+-----+-----+-----+
| 1  | VANSH JAIN | FACTORY CLERK | ROHINI WEST | 10000  | 101/1    |
| 2  | SAMEER SALMANI | FACTORY LABOUR | RITHALA    | 20000  | 102/2    |
| 3  | ROHAN MASSEY | FACTORY PEON  | ROHINI EAST | 30000  | 103/3    |
| 4  | RAHUL KUMAR  | FACTORY HEAD  | NIHAL VIHAR | 40000  | 104/4    |
| 5  | OM KAPOOR    | HR MANAGER    | SADAR BAZAAR | 50000  | 105/5    |
| 6  | SALMAN KHAN  | FINANCE MANAGER | KIRTI NAGAR | 60000  | 106/6    |
| 7  | RAJNIKANT    | ASSISTANT MANAGER | HAUSKHAS   | 70000  | 107/7    |
| 8  | SAHID KAPOOR | FACTORY LABOUR | MADIPUR    | 8000   | 108/8    |
| 9  | AMIR KHAN    | FACTORY WATCHMAN | KARAMPURA  | 9000   | 109/9    |
| 10 | SHARUK KHAN  | HR MANAGER    | PASCHIM VIHAR | 100000 | 110/10   |
+----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

Add column in above table date of joining, Experience. alter table employee add column experience date;

1. Display the Table

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> ALTER TABLE EMPLOYEE ADD COLUMN EXPERIENCE DATE;
Query OK, 10 rows affected (0.63 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM EMPLOYEE;
+----+-----+-----+-----+-----+-----+-----+
| ID | NAME      | DESG      | BRANCH    | SALARY | ADDRESS | EXPERIENCE |
+----+-----+-----+-----+-----+-----+-----+
| 1  | VANSH JAIN | FACTORY CLERK | ROHINI WEST | 10000  | 101/1    | NULL        |
| 2  | SAMEER SALMANI | FACTORY LABOUR | RITHALA    | 20000  | 102/2    | NULL        |
| 3  | ROHAN MASSEY | FACTORY PEON  | ROHINI EAST | 30000  | 103/3    | NULL        |
| 4  | RAHUL KUMAR  | FACTORY HEAD  | NIHAL VIHAR | 40000  | 104/4    | NULL        |
| 5  | OM KAPOOR    | HR MANAGER    | SADAR BAZAAR | 50000  | 105/5    | NULL        |
| 6  | SALMAN KHAN  | FINANCE MANAGER | KIRTI NAGAR | 60000  | 106/6    | NULL        |
| 7  | RAJNIKANT    | ASSISTANT MANAGER | HAUSKHAS   | 70000  | 107/7    | NULL        |
| 8  | SAHID KAPOOR | FACTORY LABOUR | MADIPUR    | 8000   | 108/8    | NULL        |
| 9  | AMIR KHAN    | FACTORY WATCHMAN | KARAMPURA  | 9000   | 109/9    | NULL        |
| 10 | SHARUK KHAN  | HR MANAGER    | PASCHIM VIHAR | 100000 | 110/10   | NULL        |
+----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

2. Complete the Table Definition

Update employee set experience='2002/09/09'

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
mysql> UPDATE EMPLOYEE SET EXPERIENCE='2002/09/09';
Query OK, 10 rows affected (0.09 sec)
Rows matched: 10 Changed: 10 Warnings: 0

mysql> SELECT * FROM EMPLOYEE;
```

ID	NAME	DESG	BRANCH	SALARY	ADDRESS	EXPERIENCE
1	VANSH JAIN	FACTORY CLERK	ROHINI WEST	10000	101/1	2002-09-09
2	SAMEER SALMANI	FACTORY LABOUR	RITHALA	20000	102/2	2002-09-09
3	ROHAN MASSEY	FACTORY PEON	ROHINI EAST	30000	103/3	2002-09-09
4	RAHUL KUMAR	FACTORY HEAD	NIHAL VIHAR	40000	104/4	2002-09-09
5	OM KAPOOR	HR MANAGER	SADAR BAZAAR	50000	105/5	2002-09-09
6	SALMAN KHAN	FINANCE MANAGER	KIRTI NAGAR	60000	106/6	2002-09-09
7	RAJNIKANT	ASSISTANT MANAGER	HAUSKHAS	70000	107/7	2002-09-09
8	SAHID KAPOOR	FACTORY LABOUR	MADIPUR	8000	108/8	2002-09-09
9	AMIR KHAN	FACTORY WATCHMAN	KARAMPURA	9000	109/9	2002-09-09
10	SHARUK KHAN	HR MANAGER	PASCHIM VIHAR	100000	110/10	2002-09-09

```
10 rows in set (0.00 sec)
```

3. Delete the Column DESG

Alter table employee drop DESG;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
mysql> ALTER TABLE EMPLOYEE DROP DESG;
Query OK, 10 rows affected (0.36 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM EMPLOYEE;
```

ID	NAME	BRANCH	SALARY	ADDRESS	EXPERIENCE
1	VANSH JAIN	ROHINI WEST	10000	101/1	2002-09-09
2	SAMEER SALMANI	RITHALA	20000	102/2	2002-09-09
3	ROHAN MASSEY	ROHINI EAST	30000	103/3	2002-09-09
4	RAHUL KUMAR	NIHAL VIHAR	40000	104/4	2002-09-09
5	OM KAPOOR	SADAR BAZAAR	50000	105/5	2002-09-09
6	SALMAN KHAN	KIRTI NAGAR	60000	106/6	2002-09-09
7	RAJNIKANT	HAUSKHAS	70000	107/7	2002-09-09
8	SAHID KAPOOR	MADIPUR	8000	108/8	2002-09-09
9	AMIR KHAN	KARAMPURA	9000	109/9	2002-09-09
10	SHARUK KHAN	PASCHIM VIHAR	100000	110/10	2002-09-09

```
10 rows in set (0.00 sec)
```

4. Find out details of employee whose salary is above 25000.

Select * from employee where salary>25000;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
mysql> SELECT * FROM EMPLOYEE WHERE SALARY>25000;
```

ID	NAME	BRANCH	SALARY	ADDRESS	EXPERIENCE
3	ROHAN MASSEY	ROHINI EAST	30000	103/3	2002-09-09
4	RAHUL KUMAR	NIHAL VIHAR	40000	104/4	2002-09-09
5	OM KAPOOR	SADAR BAZAAR	50000	105/5	2002-09-09
6	SALMAN KHAN	KIRTI NAGAR	60000	106/6	2002-09-09
7	RAJNIKANT	HAUSKHAS	70000	107/7	2002-09-09
10	SHARUK KHAN	PASCHIM VIHAR	100000	110/10	2002-09-09

```
6 rows in set (0.00 sec)
```

5. Find out details of employee order by salary.

Select * from employee order by salary

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT * FROM EMPLOYEE ORDER BY SALARY;
+-----+-----+-----+-----+-----+-----+
| ID | NAME          | BRANCH    | SALARY | ADDRESS | EXPERIENCE |
+-----+-----+-----+-----+-----+-----+
| 8 | SAHID KAPOOR  | MADIPUR   | 8000   | 108/8    | 2002-09-09 |
| 9 | AMIR KHAN     | KARAMPURA | 9000   | 109/9    | 2002-09-09 |
| 1 | VANSI JAIN     | ROHINI WEST | 10000  | 101/1    | 2002-09-09 |
| 2 | SAMEER SALMANI | RITHALA   | 20000  | 102/2    | 2002-09-09 |
| 3 | ROHAN MASSEY  | ROHINI EAST | 30000  | 103/3    | 2002-09-09 |
| 4 | RAHUL KUMAR   | NIHAL VIHAR | 40000  | 104/4    | 2002-09-09 |
| 5 | OM KAPOOR     | SADAR BAZAAR | 50000  | 105/5    | 2002-09-09 |
| 6 | SALMAN KHAN   | KIRTI NAGAR | 60000  | 106/6    | 2002-09-09 |
| 7 | RAJNIKANT     | HAUSKHAAS | 70000  | 107/7    | 2002-09-09 |
| 10 | SHARUK KHAN  | PASCHIM VIHAR | 100000 | 110/10   | 2002-09-09 |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.03 sec)
```

6. Calculate total no. of records in employee table.

Select count(id) from employee;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT COUNT(ID) FROM EMPLOYEE;
+-----+
| COUNT(ID) |
+-----+
|          10 |
+-----+
1 row in set (0.00 sec)
```

7. List Employees whose salary is between 10000 and 30000.

Select * from employee where salary between 10000 and 30000;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT * FROM EMPLOYEE WHERE SALARY BETWEEN 10000 AND 30000;
+-----+-----+-----+-----+-----+-----+
| ID | NAME          | BRANCH    | SALARY | ADDRESS | EXPERIENCE |
+-----+-----+-----+-----+-----+-----+
| 1 | VANSI JAIN     | ROHINI WEST | 10000  | 101/1    | 2002-09-09 |
| 2 | SAMEER SALMANI | RITHALA   | 20000  | 102/2    | 2002-09-09 |
| 3 | ROHAN MASSEY  | ROHINI EAST | 30000  | 103/3    | 2002-09-09 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

QUESTION 7: Create the following table and perform SQL commands. Student

(Roll_no,name, age, course, marks).

1. List all those students who are greater than 18 years of age and have opted for MBA

course.

2. List name of student whose name end with 'i'.

3. Find out total number of records in table.

4. Find out the name, course, marks and sort in the order of marks.

5. Display name and course of student.

6. Find the Student with Max marks.

7. List the name of students ORDER BY Roll_no in Descending order.

8. Find out the Average of all the Marks. Display it as Average_Marks.

a) Create database students;

b) Use students;

c) Create table student(roll_no int, name varchar (20), age int, course varchar(10), marks int);

1) Insert Into student values(01 'Saktiman', 20, 'Mba', 99);

2) Insert Into student values(02, 'Spider man', 19, 'Bba', 80);

3) Insert Into student values(03, 'Padman', 20, 'Mba', 85);

4) Insert Into student values(04, 'Rockey bhai', 21, 'Bcom', 75);

5) Insert Into student values(05, 'Pushpa jn', 18, 'Bba', 60);

select * from student;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT * FROM STUDENT;
+-----+-----+-----+-----+-----+
| ROLL_NO | NAME      | AGE | COURSE | MARKS |
+-----+-----+-----+-----+-----+
| 1       | SAKTIMAN  | 20  | MBA    | 99    |
| 2       | SPIDER MAN | 19  | BBA    | 80    |
| 3       | PADMAN    | 20  | MBA    | 85    |
| 4       | ROCKEY BHAI | 21  | BCOM   | 75    |
| 4       | PUSHPA JN | 18  | BBA    | 60    |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

1. List all those students who are greater than 18 years of age and have opted for MBA course.

Select * from student where age>18 and course= 'mba';

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT * FROM STUDENT WHERE AGE>18 AND COURSE= 'MBA';
+-----+-----+-----+-----+-----+
| ROLL_NO | NAME      | AGE | COURSE | MARKS |
+-----+-----+-----+-----+-----+
| 1       | SAKTIMAN  | 20  | MBA    | 99    |
| 3       | PADMAN    | 20  | MBA    | 85    |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
```

a) List name of student whose name end with 'R%'.

Select * from student where name like 'i%';

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT * FROM STUDENT WHERE NAME LIKE 'R%';
+-----+-----+-----+-----+-----+
| ROLL_NO | NAME      | AGE | COURSE | MARKS |
+-----+-----+-----+-----+-----+
| 4       | ROCKEY BHAI | 21  | BCOM   | 75    |
+-----+-----+-----+-----+-----+
1 row in set (0.08 sec)

mysql>
```

b) Find out total number of records in table.

select count(roll_no) from student;


```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT COUNT(ROLL_NO) FROM STUDENT;
+-----+
| COUNT(ROLL_NO) |
+-----+
|          5      |
+-----+
1 row in set (0.09 sec)

mysql>
```

c) Find out the name, course, marks and sort in the order of marks.

Select name, course, marks from student order by marks;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT NAME, COURSE, MARKS FROM STUDENT ORDER BY MARKS;
+-----+-----+-----+
| NAME      | COURSE | MARKS |
+-----+-----+-----+
| PUSHPA JN | BBA    | 60    |
| ROCKEY BHAI | BCOM  | 75    |
| SPIDER MAN | BBA    | 80    |
| PADMAN    | MBA    | 85    |
| SAKTIMAN  | MBA    | 99    |
+-----+-----+-----+
5 rows in set (0.09 sec)

mysql>
```

d) Display name and course of student.

Select name, course from student

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT NAME, COURSE FROM STUDENT;
+-----+-----+
| NAME      | COURSE |
+-----+-----+
| SAKTIMAN  | MBA    |
| SPIDER MAN | BBA    |
| PADMAN    | MBA    |
| ROCKEY BHAI | BCOM  |
| PUSHPA JN | BBA    |
+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

e) Find the Student with Max marks.

Select max(marks), name from student;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT MAX(MARKS), NAME FROM STUDENT;
+-----+-----+
| MAX(MARKS) | NAME      |
+-----+-----+
|          99 | SAKTIMAN  |
+-----+-----+
1 row in set (0.00 sec)

mysql>
```

f) List the name of students ORDER BY Roll_no in Descending order.

Select name from student order by roll_no desc;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT NAME FROM STUDENT ORDER BY ROLL_NO DESC;
+-----+
| NAME      |
+-----+
| ROCKEY BHAI |
| PUSHPA JN  |
| PADMAN     |
| SPIDER MAN  |
| SAKTIMAN    |
+-----+
5 rows in set (0.00 sec)

mysql>
```

g) Find out the Average of all the Marks. Display it as Average_Marks

Select avg(marks) from student;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT AVG(MARKS) FROM STUDENT;
+-----+
| AVG(MARKS) |
+-----+
|    79.8000 |
+-----+
1 row in set (0.09 sec)

mysql>
```

Question8: Create the Following Tables. Insert at least 10 records in each.

Table1: Supplier (S_No, Sname, Status, City) Table2: Parts (P_No, Pname, Color, Weight, City) Table 3: SP (S_No, P_No, Quantity)

Answer the following queries in SQL:

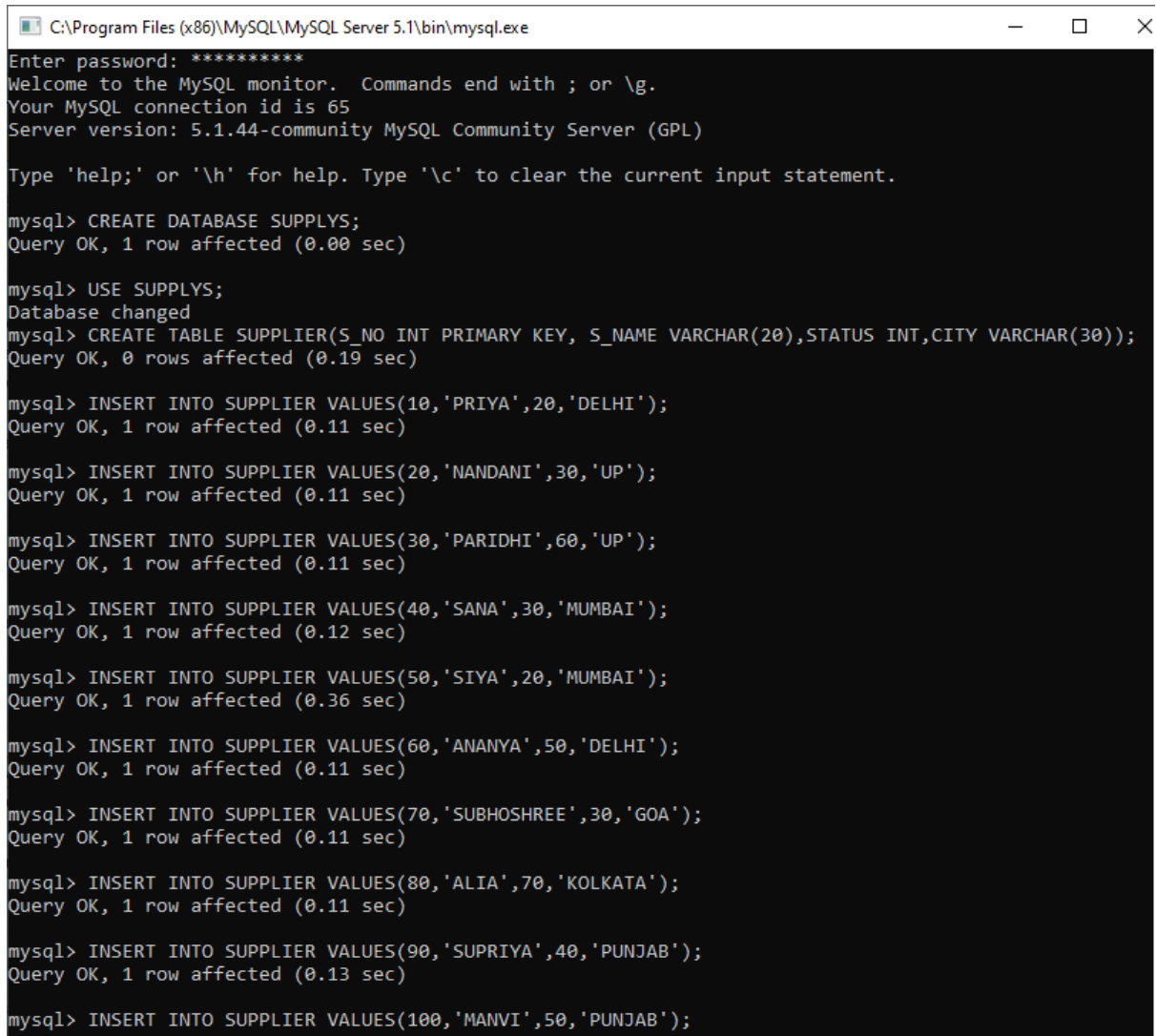
- a) Find the supplier for city = 'Delhi'**
- b) Find suppliers whose name start with 'AB'**
- c) Find all suppliers whose status is 10, 20 or 30**
- d) Find total number of city of all suppliers**
- e) Find the name of suppliers who supplies quantity of the item P1 more than 50**

Answer

Table 1

- a) Create database dat;
- b) Use dat;
- c) Create table supplier (s_no int primary key, s_name varchar(20), status int, city varchar(30));
 - 1) Insert Into supplier values(10, 'Priya', 20, 'delhi');
 - 2) Insert Into supplier values(20, 'Nandani', 30, 'up');
 - 3) Insert Into supplier values(30, 'Paridhi', 60, 'up');
 - 4) Insert Into supplier values(40, 'Sana', 30, 'mumbai');
 - 5) Insert Into supplier values(50, 'Siya', 20, 'mumbai');
 - 6) Insert Into supplier values(60, 'Ananya', 50, 'delhi');
 - 7) Insert Into supplier values(70, 'Subhoshree', 30, 'goa');
 - 8) Insert Into supplier values(80, 'Alia', 70, 'up');

9) Insert Into supplier values(90, 'Supriya', 40, 'punjab');
10) Insert Into supplier values(100, 'Manvi', 50, 'punjab');
Select * from supplier;



```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 65
Server version: 5.1.44-community MySQL Community Server (GPL)

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE SUPPLYS;
Query OK, 1 row affected (0.00 sec)

mysql> USE SUPPLYS;
Database changed
mysql> CREATE TABLE SUPPLIER(S_NO INT PRIMARY KEY, S_NAME VARCHAR(20),STATUS INT,CITY VARCHAR(30));
Query OK, 0 rows affected (0.19 sec)

mysql> INSERT INTO SUPPLIER VALUES(10,'PRIYA',20,'DELHI');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO SUPPLIER VALUES(20,'NANDANI',30,'UP');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO SUPPLIER VALUES(30,'PARIDHI',60,'UP');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO SUPPLIER VALUES(40,'SANA',30,'MUMBAI');
Query OK, 1 row affected (0.12 sec)

mysql> INSERT INTO SUPPLIER VALUES(50,'SIYA',20,'MUMBAI');
Query OK, 1 row affected (0.36 sec)

mysql> INSERT INTO SUPPLIER VALUES(60,'ANANYA',50,'DELHI');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO SUPPLIER VALUES(70,'SUBHOSHREE',30,'GOA');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO SUPPLIER VALUES(80,'ALIA',70,'KOLKATA');
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO SUPPLIER VALUES(90,'SUPRIYA',40,'PUNJAB');
Query OK, 1 row affected (0.13 sec)

mysql> INSERT INTO SUPPLIER VALUES(100,'MANVI',50,'PUNJAB');
```

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT * FROM SUPPLIER;
+-----+-----+-----+-----+
| S_NO | S_NAME | STATUS | CITY |
+-----+-----+-----+-----+
| 10 | PRIYA | 20 | DELHI |
| 20 | NANDANI | 30 | UP |
| 30 | PARIDHI | 60 | UP |
| 40 | SANA | 30 | MUMBAI |
| 50 | SIYA | 20 | MUMBAI |
| 60 | ANANYA | 50 | DELHI |
| 70 | SUBHOSHREE | 30 | GOA |
| 80 | ALIA | 70 | KOLKATA |
| 90 | SUPRIYA | 40 | PUNJAB |
| 100 | MANVI | 50 | PUNJAB |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

Table 2

a) Create table parts(p_no varchar(20) primary key, p_name varchar(20), color varchar(10), weight int,city varchar(30));

a. Insert Into parts values('p1', 'Printer', 'white', 400, 'delhi');

b. Insert Into parts values('p2', 'Speaker', 'white', 500, 'mumbai');

c. Insert Into parts values('p3', 'Keyboard', 'white', 500, 'delhi');

d. Insert Into parts values('p4', 'Mouse', 'white', 100, 'delhi');

e. Insert Into parts values('p5', 'Cpu', 'black', 400, 'mumbai');

f. Insert Into parts values('p6', 'Cpu', 'black', 900, 'delhi');

g. Insert Into parts values('p7', 'Speaker', 'black', 400, 'mumbai');

h. Insert Into parts values('p8', 'Monitor', 'black', 1000, 'punjab');

i. Insert Into parts values('p9', 'Ups', 'blue', 700, 'mumbai');

j. Insert Into parts values('p10', 'Webcam', 'brown', 200, 'mathura');

Select * from parts;

```

C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
mysql> CREATE TABLE PARTS(P_NO VARCHAR(20) PRIMARY KEY, P_NAME VARCHAR(20), COLOUR VARCHAR(10),WEIGHT
INT,CITY VARCHAR(30));
Query OK, 0 rows affected (0.16 sec)

mysql> INSERT INTO PARTS VALUES('P1','PRINTER','WHITE',400,'DELHI');
Query OK, 1 row affected (0.02 sec)

mysql> INSERT INTO PARTS VALUES('P2','SPEAKER','WHITE',500,'MUMBAI');
Query OK, 1 row affected (0.13 sec)

mysql> INSERT INTO PARTS VALUES('P3','KEYBOARD','WHITE',500,'DELHI');
Query OK, 1 row affected (0.13 sec)

mysql> INSERT INTO PARTS VALUES('P4','MOUSE','WHITE',100,'DELHI');
Query OK, 1 row affected (0.12 sec)

mysql> INSERT INTO PARTS VALUES('P5','CPU','BLACK',400,'MUMBAI');
Query OK, 1 row affected (0.13 sec)

mysql> INSERT INTO PARTS VALUES('P6','CPU','BLACK',900,'DELHI');
Query OK, 1 row affected (0.47 sec)

mysql> INSERT INTO PARTS VALUES('P7','SPEAKER','BLACK',400,'MUMBAI');
Query OK, 1 row affected (0.13 sec)

mysql> INSERT INTO PARTS VALUES('P8','MONITOR','BLACK',1000,'PUNJAB');
Query OK, 1 row affected (0.13 sec)

mysql> INSERT INTO PARTS VALUES('P9','UPS','BLUE',700,'MUMBAI');
Query OK, 1 row affected (0.13 sec)

mysql> INSERT INTO PARTS VALUES('P10','WEBCAM','BROWN',200,'MATHURA');
Query OK, 1 row affected (0.13 sec)

```

```

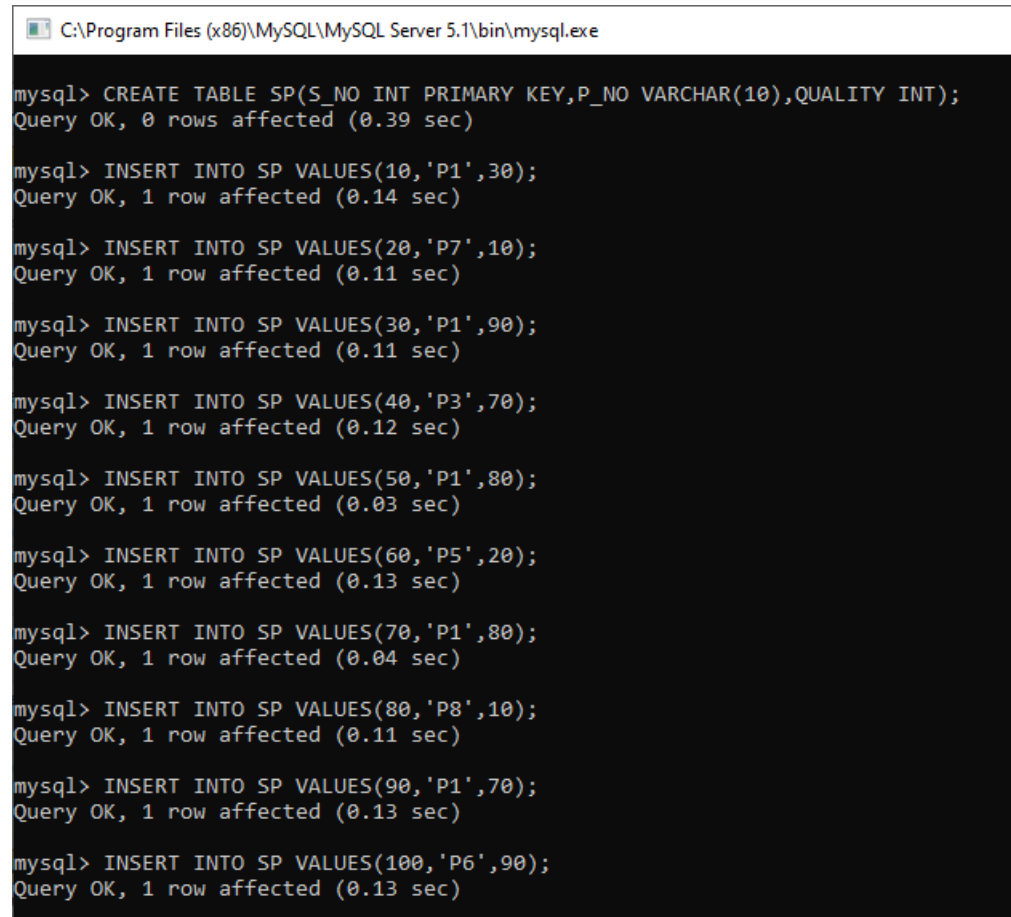
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
mysql> SELECT * FROM PARTS;
+-----+-----+-----+-----+-----+
| P_NO | P_NAME | COLOUR | WEIGHT | CITY |
+-----+-----+-----+-----+-----+
| P1   | PRINTER | WHITE  | 400    | DELHI |
| P10  | WEBCAM  | BROWN  | 200    | MATHURA |
| P2   | SPEAKER | WHITE  | 500    | MUMBAI |
| P3   | KEYBOARD | WHITE  | 500    | DELHI |
| P4   | MOUSE   | WHITE  | 100    | DELHI |
| P5   | CPU     | BLACK  | 400    | MUMBAI |
| P6   | CPU     | BLACK  | 900    | DELHI |
| P7   | SPEAKER | BLACK  | 400    | MUMBAI |
| P8   | MONITOR | BLACK  | 1000   | PUNJAB |
| P9   | UPS     | BLUE   | 700    | MUMBAI |
+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

```

Table 3

- a) Create table sp(s_no int primary key, p_no varchar(10), quantity int);
- 1) Insert Into sp values(10, 'p1', 30);
 - 2) Insert Into sp values(20, 'p7', 10);
 - 3) Insert Into sp values(30, 'p1', 90);

4) Insert Into sp values(40, 'p3', 70);
5) Insert Into sp values(50, 'p1', 80);
6) Insert Into sp values(60, 'p5', 20);
7) Insert Into sp values(70, 'p1', 80);
8) Insert Into sp values(80, 'p8', 10);
9) Insert Into sp values(90, 'p1', 70);
10) Insert Into sp values(100, 'p6', 90);
Select * from sp;



```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> CREATE TABLE SP(S_NO INT PRIMARY KEY,P_NO VARCHAR(10),QUALITY INT);
Query OK, 0 rows affected (0.39 sec)

mysql> INSERT INTO SP VALUES(10,'P1',30);
Query OK, 1 row affected (0.14 sec)

mysql> INSERT INTO SP VALUES(20,'P7',10);
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO SP VALUES(30,'P1',90);
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO SP VALUES(40,'P3',70);
Query OK, 1 row affected (0.12 sec)

mysql> INSERT INTO SP VALUES(50,'P1',80);
Query OK, 1 row affected (0.03 sec)

mysql> INSERT INTO SP VALUES(60,'P5',20);
Query OK, 1 row affected (0.13 sec)

mysql> INSERT INTO SP VALUES(70,'P1',80);
Query OK, 1 row affected (0.04 sec)

mysql> INSERT INTO SP VALUES(80,'P8',10);
Query OK, 1 row affected (0.11 sec)

mysql> INSERT INTO SP VALUES(90,'P1',70);
Query OK, 1 row affected (0.13 sec)

mysql> INSERT INTO SP VALUES(100,'P6',90);
Query OK, 1 row affected (0.13 sec)
```

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT * FROM SP;
+-----+-----+-----+
| S_NO | P_NO | QUALITY |
+-----+-----+-----+
| 10   | P1   | 30      |
| 20   | P7   | 10      |
| 30   | P1   | 90      |
| 40   | P3   | 70      |
| 50   | P1   | 80      |
| 60   | P5   | 20      |
| 70   | P1   | 80      |
| 80   | P8   | 10      |
| 90   | P1   | 70      |
| 100  | P6   | 90      |
+-----+-----+-----+
10 rows in set (0.00 sec)
```

1) Find the supplier for city = 'Delhi'

Select s_name from supplier where city='delhi';

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT S_NAME FROM SUPPLIER WHERE CITY='DELHI';
+-----+
| S_NAME |
+-----+
| PRIYA  |
| ANANYA |
+-----+
2 rows in set (0.00 sec)
```

2) Find suppliers whose name start with 'PA'

Select * from supplier where s_name like 'pa%';

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT * FROM SUPPLIER WHERE S_NAME LIKE 'PA%';
+-----+-----+-----+-----+
| S_NO | S_NAME | STATUS | CITY |
+-----+-----+-----+-----+
| 30   | PARIDHI | 60     | UP   |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

3) Find total number of city of all suppliers

Select count(city) from supplier;


```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT COUNT(CITY) FROM SUPPLIER;
+-----+
| COUNT(CITY) |
+-----+
|          10 |
+-----+
1 row in set (0.00 sec)
```

4. Find all suppliers whose status is 10, 20 or 30

Select s_name from supplier where status in (10,20,30)

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT S_NAME FROM SUPPLIER WHERE STATUS IN (10,20,30);
+-----+
| S_NAME |
+-----+
| PRIYA  |
| NANDANI |
| SANA   |
| SIYA   |
| SUBHOSHREE |
+-----+
5 rows in set (0.00 sec)
```

5. Find the name of suppliers who supplies quantity of the item P1 more than 40

Select s_name from supplier, sp where supplier.s_no=sp.s_no and sp.p_no='p1' and sp.quantity>40;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> SELECT S_NAME FROM SUPPLIER WHERE STATUS>40;
+-----+
| S_NAME |
+-----+
| PARIDHI |
| ANANYA  |
| ALIA    |
| MANVI   |
+-----+
4 rows in set (0.01 sec)

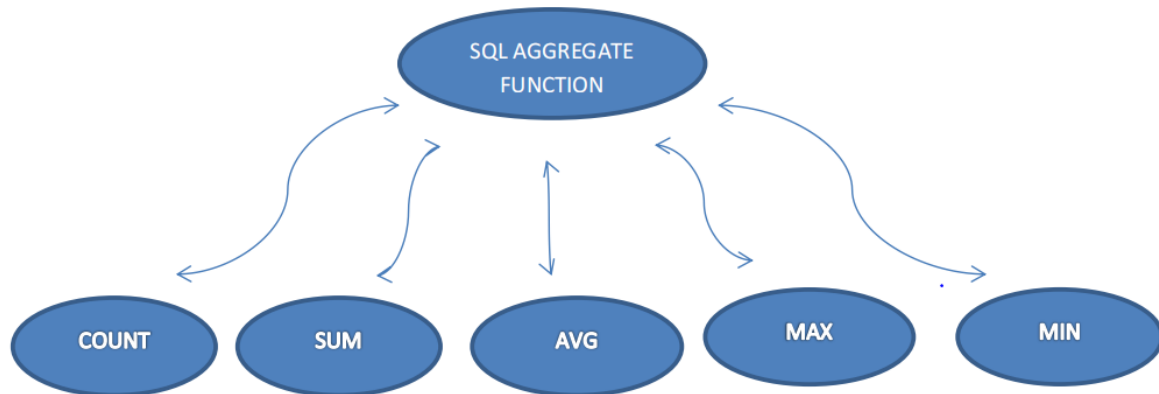
mysql>
```

QUESTION 9: Write a command Aggregate Functions?

ANSWER

SQL aggregation function is used to perform the calculations on multiple rows of a single column of a table. It returns a single value. It is also used to summarize the data.

Types of SQL Aggregation Function



COUNT FUNCTION

1. COUNT function is used to Count the number of rows in a database table. It can work on both numeric and non-numeric data types.
2. COUNT function uses the COUNT (*) that returns the count of all the rows in a specified table. COUNT (*) considers duplicate and Null.

Syntax: -

COUNT (*) or COUNT ([ALL|DISTINCT] expression

EXAMPLE:-

USE OF SYNTAX ARE AS FOLLOW:-

1. SELECT COUNT(STUDENT_NAME) FROM STUDENT;
2. SELECT COUNT(STUDENT_MARKS) FROM STUDENT;

```

C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
mysql> SELECT COUNT(STUDENT_NAME) FROM STUDENT;
+-----+
| COUNT(STUDENT_NAME) |
+-----+
|          8          |
+-----+
1 row in set (0.00 sec)

mysql> SELECT COUNT(STUDENT_MARKS) FROM STUDENT;
+-----+
| COUNT(STUDENT_MARKS) |
+-----+
|          8          |
+-----+
1 row in set (0.00 sec)

```

SUM FUNCTION

Sum function is used to calculate the sum of all selected columns. It works on numeric fields only.

Syntax:-

SUM () or SUM ([ALL|DISTINCT] expression): -

EXAMPLE:-

USE OF SYNTAX ARE AS FOLLOW: -

1. SELECT SUM(STUDENT_ROLLNO) FROM STUDENT;
2. SELECT SUM(STUDENT_MARKS) FROM STUDENT;

```

C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
mysql> SELECT SUM(STUDENT_ROLLNO) FROM STUDENT;
+-----+
| SUM(STUDENT_ROLLNO) |
+-----+
|          3916        |
+-----+
1 row in set (0.00 sec)

mysql> SELECT SUM(STUDENT_MARKS) FROM STUDENT;
+-----+
| SUM(STUDENT_MARKS) |
+-----+
|          626        |
+-----+
1 row in set (0.00 sec)

```

AVG FUNCTION

The AVG function is used to calculate the average value of the numeric type. AVG function returns the average of all non-Null values.

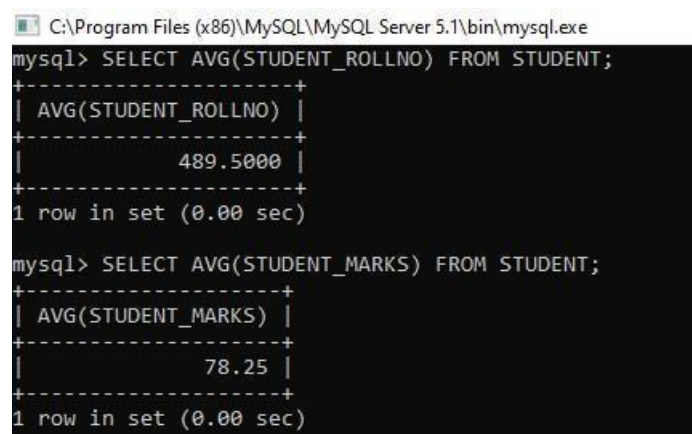
Syntax:-

AVG() or AVG([ALL|DISTINCT] expression)

EXAMPLE:-

USE OF SYNTAX ARE AS FOLLOW: -

1. SELECT AVG(STUDENT_ROLLNO) FROM STUDENT;
2. SELECT AVG(STUDENT_MARKS) FROM STUDENT;



```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
mysql> SELECT AVG(STUDENT_ROLLNO) FROM STUDENT;
+-----+
| AVG(STUDENT_ROLLNO) |
+-----+
|          489.5000 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT AVG(STUDENT_MARKS) FROM STUDENT;
+-----+
| AVG(STUDENT_MARKS) |
+-----+
|           78.25 |
+-----+
1 row in set (0.00 sec)
```

MAX FUNCTION

MAX function is used to find the maximum value of a certain column. This function determines the largest value of all selected values of a column.

Syntax:-

MAX() or MAX([ALL|DISTINCT] expression)

EXAMPLE:-

USE OF SYNTAX ARE AS FOLLOW:-

1. SELECT MAX(STUDENT_ROLLNO) FROM STUDENT;
2. SELECT MAX(STUDENT_MARKS) FROM STUDENT;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
mysql> SELECT MAX(STUDENT_ROLLNO) FROM STUDENT;
+-----+
| MAX(STUDENT_ROLLNO) |
+-----+
|          889        |
+-----+
1 row in set (0.00 sec)

mysql> SELECT MAX(STUDENT_MARKS) FROM STUDENT;
+-----+
| MAX(STUDENT_MARKS) |
+-----+
|          99         |
+-----+
1 row in set (0.01 sec)
```

MIN FUNCTION

MIN function is used to find the minimum value of a certain column. This function determines the smallest value of all selected values of a column.

Syntax:-

MIN() or MIN([ALL|DISTINCT] expression)

EXAMPLE:-

USE OF SYNTAX ARE AS FOLLOW:-

1. SELECT MIN(STUDENT_ROLLNO) FROM STUDENT;
2. SELECT MIN(STUDENT_MARKS) FROM STUDENT;

```
C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe
mysql> SELECT MIN(STUDENT_ROLLNO) FROM STUDENT;
+-----+
| MIN(STUDENT_ROLLNO) |
+-----+
|          211        |
+-----+
1 row in set (0.00 sec)

mysql> SELECT MIN(STUDENT_MARKS) FROM STUDENT;
+-----+
| MIN(STUDENT_MARKS) |
+-----+
|          45         |
+-----+
1 row in set (0.00 sec)
```

Question 10: What do you understand by ER Modelling? Explain all the symbols of ER Model with examples.

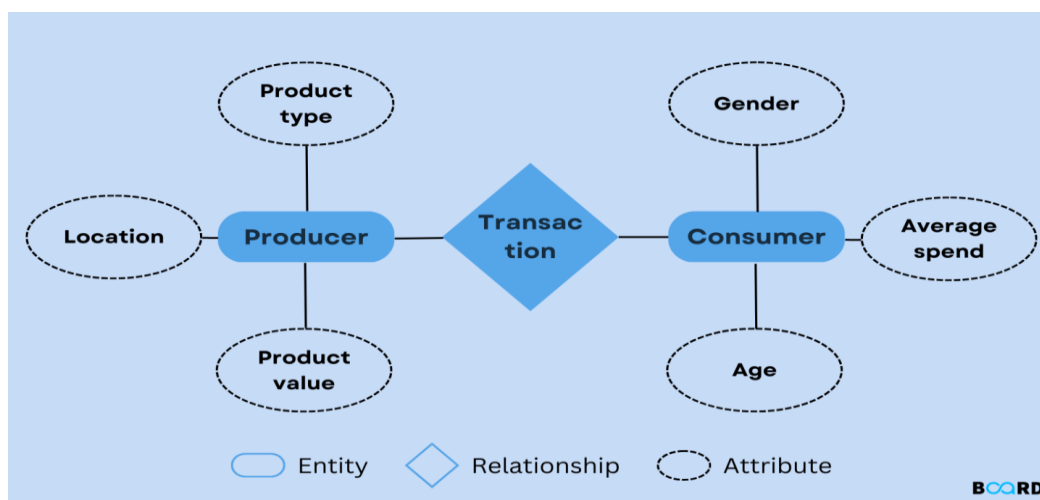
An entity relationship diagram (ERD), also known as an entity relationship model, is a graphical representation of an information system that depicts the relationships among people, objects, places, concepts or events within that system. An ERD is a data modelling technique that can help define business processes and be used as the foundation for a relational database. There are three basic components of an entity relationship diagram:

1. Entities, which are objects or concepts that can have data stored about them.
2. Attributes, which are properties or characteristics of entities. An ERD attribute can be denoted as a primary key, which identifies a unique attribute, or a foreign key, which can be assigned to multiple attributes.
3. The relationships between and among those entities. An entity relationship diagram showing relationships between sales reps, customers and product orders. For example, an ERD representing the information system for a company's sales department might start with graphical representations of entities such as the sales representative, the customer, the customer's address, the customer's order, the product and the warehouse. Then lines or other symbols can be used to represent the relationship between entities, and text can be used to label the relationships. A cardinality notation can then define the attributes of the relationship between the entities. Cardinalities can denote that an entity is optional (for example, a sales rep could have no customers or could have many) or mandatory (for example, there must be at least one product listed in an order.) The three main cardinalities are:
 1. A one-to-one relationship (1:1). For example, if each customer in a database is associated with one mailing address.

2. A one-to-many relationship (1: M). For example, a single customer might place an order for multiple products. The customer is associated with multiple entities, but all those entities have a single connection back to the same customer.³⁷

4 A many-to-many relationship (M: N). For example, at a company where all call centre agents work with multiple customers, each agent is associated with multiple customers, and multiple customers might also be associated with multiple agents.

ER Diagrams Symbols, and notations



ER Diagram Symbols and Notations

Components of an E-R diagram

An E-R diagram constitutes of following Components

- A. Entity: - Any real-world object can be represented as an entity about which data can be stored in a database. All the real-world objects like a book, an organization, a product, a car, a person are the examples of an entity. Any living or non-living objects can be represented by an entity. An entity is symbolically represented by a rectangle enclosing its name.

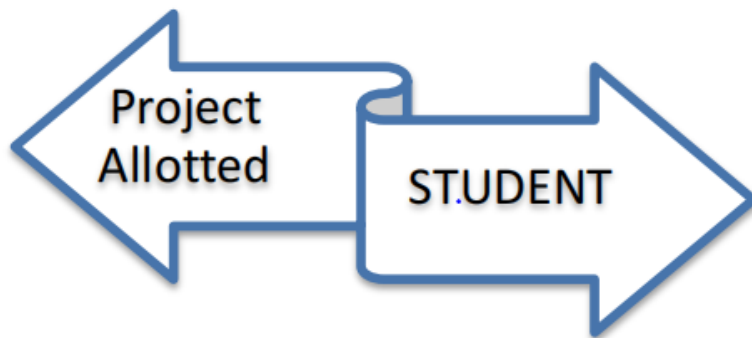


Entities can be characterized into two types:

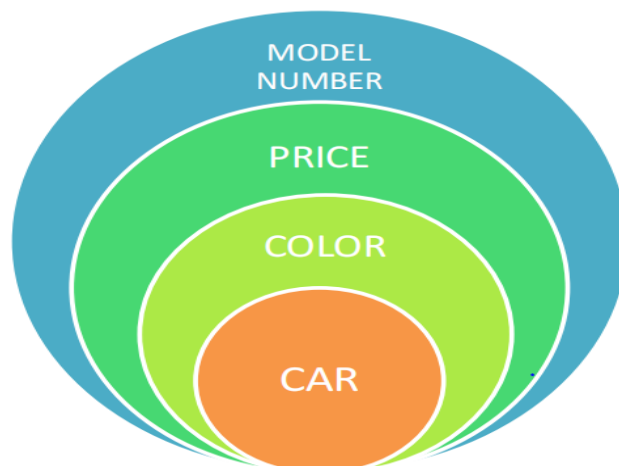
- Strong entity: A strong entity has a primary key attribute which uniquely identifies each entity. Symbol of strong entity is same as an entity.



- Weak entity: A weak entity does not have a primary key attribute and depends on other entity via a foreign key attribute.

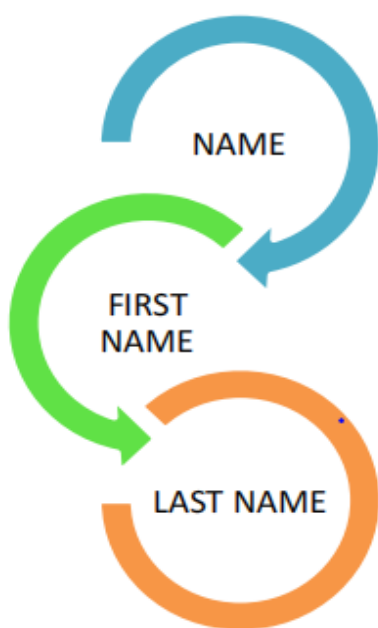


- B. Attribute: - Each entity has a set of properties. These properties of each entity are termed as attributes. For example, a car entity would be described by attributes such as price, registration number, model number, color etc. Attributes are indicated by ovals in an e-r diagram.



A primary key attribute is depicted by an underline in the e-r diagram. An attribute can be characterized into following types:

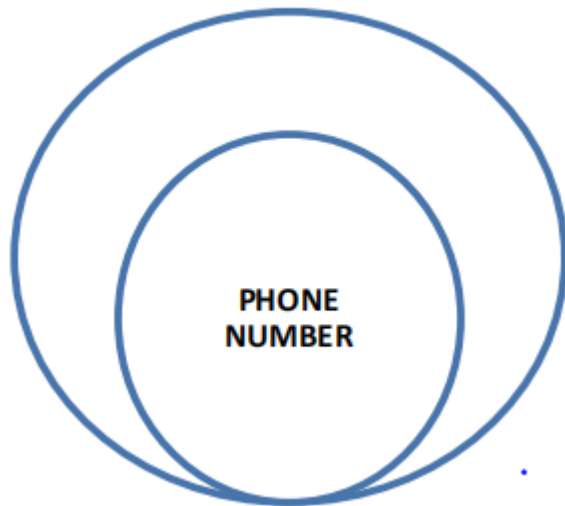
- Simple attribute: - An attribute is classified as a simple attribute if it cannot be partitioned into smaller components. For example, age and sex of a person. A simple attribute is represented by an oval.
- Composite attribute: - A composite attribute can be subdivided into smaller components which further form attributes. For example, 'name' attribute of an entity "person" can be broken down into first name and last name which further form attributes. Grouping of these related attributes forms a composite attribute. 'name' is the composite attribute in this exam



• Composite Attribute

Single valued attribute: - If an attribute of a particular entity represents single value for each instance, then it is called a single-valued attribute. For example, Ramesh, Kamal and Suraj are the instances of entity 'student' and each of them is issued a separate roll number. A single oval is used to represent this attribute.

- Multi valued attribute: – An attribute which can hold more than one value, it is then termed as multi-valued attribute. For example, phone number of a person. Symbol of multi-valued attribute is shown below.



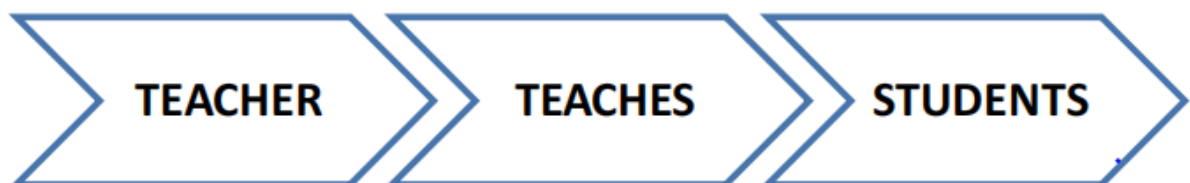
- **Multi Valued Attribute**

Derived attribute: A derived attribute calculates its value from another attribute. For example, 'age' is a derived attribute if it calculates its value from 'current date' & 'birth date' attributes. A derived attribute is represented by a dashed oval.

C. Relationships: - A relationship is defined as bond or attachment between 2 or more entities. Normally, a verb in sentence signifies a relationship. For example,

- o an employee assigned a project.
- o Teacher teaches a student.
- o Author writes a book.

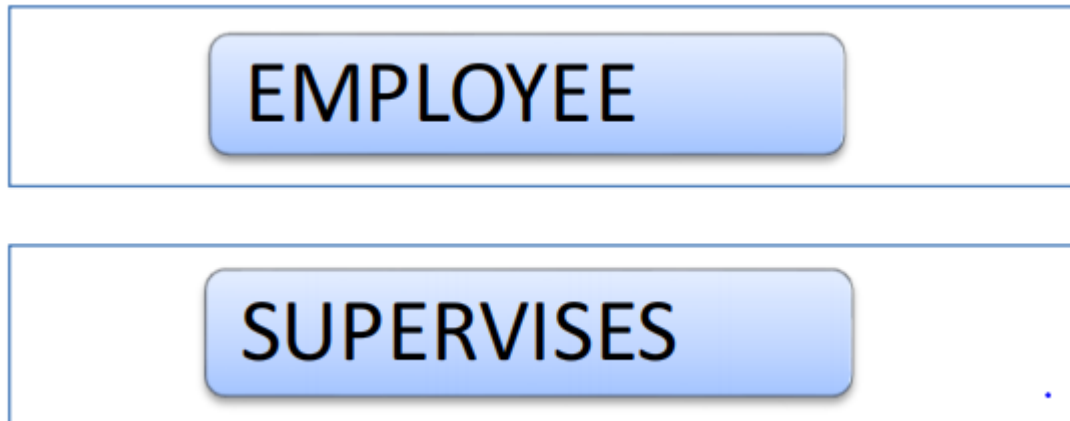
A diamond is used to symbolically represent a relationship in the e-r diagram.



RELATIONSHIP

a). **Degree of relationship:** - It signifies the number of entities involved in a relationship. Degree of a relationship can be classified into following types:

- Unary relationship: - If only single entity is involved in a relationship, then it is a unary relationship. For example, an employee(manager) supervises another employee.



- Binary relationships: - when two entities are associated to form a relation, then it is known as a

binary relationship. For example, A person works in a company. Most of the times we use only binary

relationship in an e-r diagram. The teacher-student example shown above signifies a binary relationship.

Other types of relationships are ternary and quaternary. As the name signifies, a ternary relationship is

associated with three entities and a quaternary relationship is associated with four entities.

b.) Connectivity of a relationship: - Connectivity of a relationship describes, how many instances of one entity type are linked to how many instances of another entity type. Various categories of connectivity of a relationship are;

- One to One (1:1) – “Student allotted a project” signifies a one-to-one relationship because only one instance of an entity is related with exactly one instance of another entity type.



ONE TO ONE

- One to Many (1:M) – “A department recruits’ faculty” is a one-to-many relationship because a department can recruit more than one faculty, but a faculty member is related to only one department.



- Many to One (M:1) – “Many houses are owned by a person” is a many-to-one relationship because a person can own many houses but a particular house is owned only a person.
- Many to Many (M: N) – “Author writes books” is a many-to-many relationship because an author can write many books and a book can be written by many authors.



Question 11: Draw Entity Relationship diagram of an Organization.

Entity: A definable things such as person, object, concept or event that can have data stored about it. Think of entities as noun.

In this ERD of organization we have three Entities: - Department, Project, Employee.

Attributes: A property or characteristic of an entity. Often shown as an oval.

In this ERD of organization: -

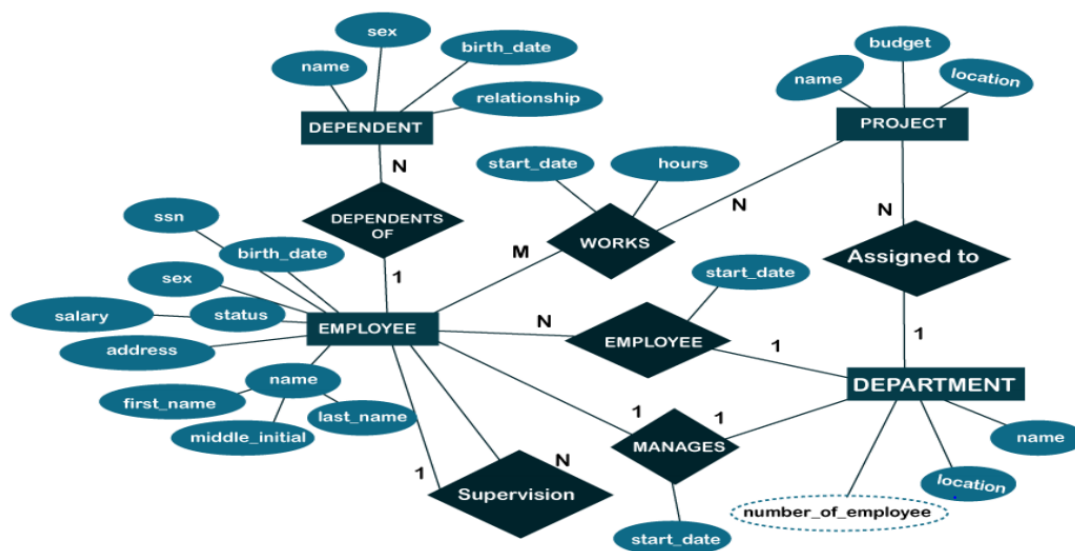
1 Department has: - Name, D_no, Location

2 Project has: - P_Name, P_Duration, P_No.

3 Employee: -Name, DOB, Gender, Phone_No. Email.

Relationship: - Relationships are typically shown as diamonds on the connecting lines. In this ERD of organization we have four relationships: - works for, supervise, control.

PRIMARY KEY: - The field that is unique for all the record occurrences.it denotes as Underline. In this ERD we have D_no in as primary key in Department, P_no in Project and Email in Employee.



Question 12: Draw Entity Relationship diagram of a Banking System.

Entity: A definable things such as person, object, concept or event that can have data stored about it think of entities as noun.

In this ERD of organization we have four Entities: - Bank, Card facility, Account, Customer.

TRANSACTION, Accounts.

Attributes: A property or characteristic of an entity. Often shown as an oval.

In this ERD of organization: -

a. BANK: - B_id, Name,

Address

b. Account: - Acc_no, Acc_type,

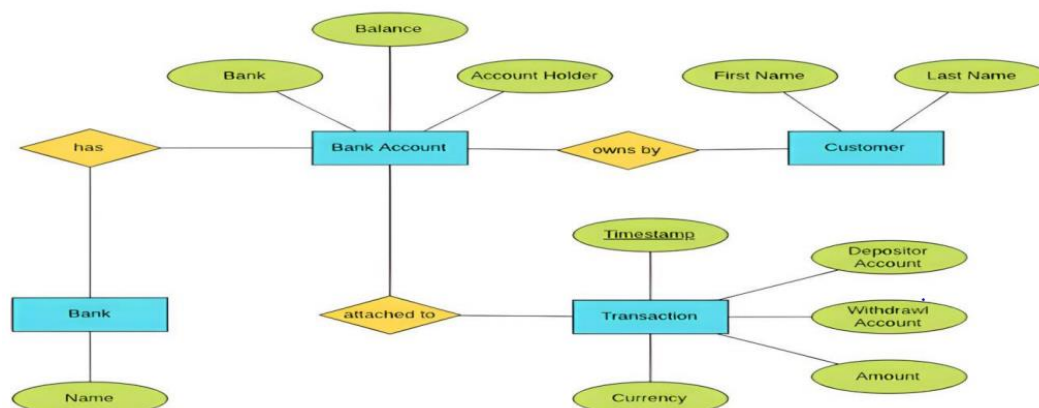
balance.

c. Customer: - Name, C_Id, Phone.no., Address.

d. Card facilities: - Card_no., Credit_card, Debit_card, Rupee-pay

Relationship: - Relationships are typically shown as diamonds on the connecting lines. In this ERD of organization we have four relationships:- Offers, Has, Has.

PRIMARY KEY: - The field that is unique for all the record occurrences.it denotes as Underline. In this ERD we have bank_id in as primary key in Bank, Acc_No in Account, C_id in Customer, Card_No. in Card facilities.



Question 13: Draw Entity Relationship diagram of a Hospital Management System.

Entity: A definable things such as person, Object, concept or event that can have data stored about it think of entities as noun.

In this ERD we have five Entities: - Patient, Medical records, Hospital, Doctor, Attributes:

A property or characteristic of an entity. Often shown as an oval.

In this ERD: -

i. Patient: - P. Name, P.Id, P. Address, P.Diagnosis

ii. Hospital:- Hos.Id, Hos.Name, Hos.No.

iii. Medical records: - R.No, Room. Charge, Doc.

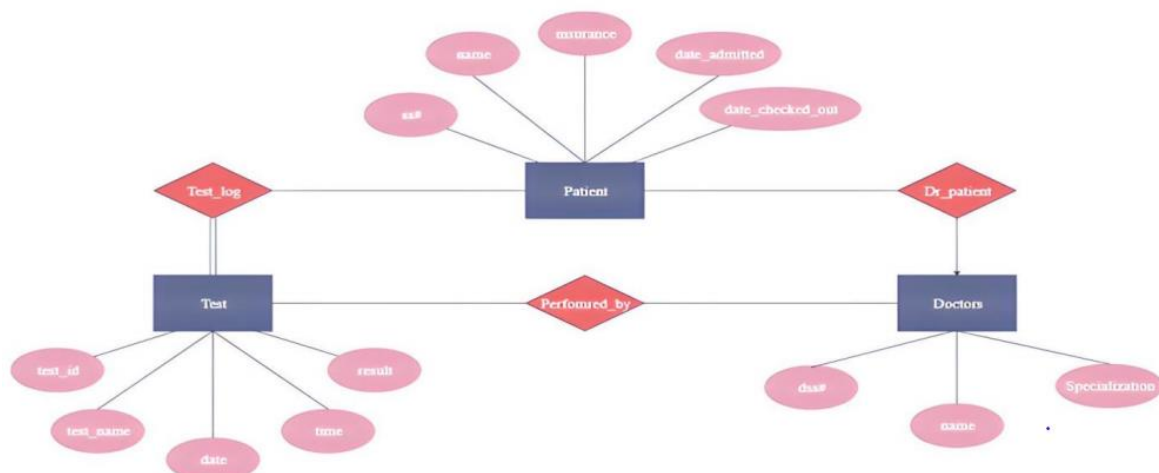
iv. Charge Doctor:- Doc_id,Doc_Name, Salary, Qualification

Relationship: - Relationships are typically shown as diamonds on the connecting lines. In this ERD of organization we have four relationships: - Has, admitted in, has

PRIMARY KEY: - The field that is unique for all the record occurrences.it denotes as Underline. In this

ERD we have P-id in patient, Hos-id in Hospital, R.No in Medical record, Doc-id in Doctor.

ER diagram of Hospital



Question 14: Draw Entity Relationship diagram of a University.

Entity: A definable things such as person, Object, concept or event that can have data stored about it think of entities as noun. In this ERD we have five Entities:

- Student, University, College, Course and Faculty Attributes:

A property or characteristic of an entity. Often shown as an oval.

In this ERD:-

- a) Student: - Stu_Id, Stu_Name, DOB, Phone_no
- b) University: - Uni_name, Uni_id, Location, Fees
- c) College: - Name, Location, Clg_id
- d) Course: - Course_Id, Name, Duration
- e) Faculty: - Salary, Fac_name, Fac_id, Qualification

Relationship: - Relationships are typically shown as diamonds on the connecting lines. In this ERD of organization we have four relationship: -Has, Study in, Enrols in, Teaches by

PRIMARY KEY: - The field that is unique for all the record occurrences.it denotes as Underline. In this ERD we have Stu_id In Student, Uni_id in University, Clg_id in College, Course_id in Course, Fac_name in Faculty.

